

Cost-Benefit Memo: Model Complexity for ED Triage

To: Dr De Freitas · ED Board · Martina Griffith (Clinical IT Lead)

From: Aiden Daniel · CariSurg MedTech Pathways · Mercer General Hospital ED Triage Feasibility Pilot

Verdict

Recommend Gradient Boosting as the Phase 3 development track — it matches the best clinical benefit available this week (recall on critical, ESI 1, patients) at roughly 1/140th the training cost of the next-best option. None of this week's models are yet safe to deploy, and that has to be said plainly before anything else in this memo.

Dataset & Methods Recap

Same dataset and split as the Week 6 final report: the Yale EMLC ED triage extract (55,121 visits), 80/20 stratified split on ESI, fixed random_state = 42. This week added engineered clinical features on top of the Week 6 feature set — a shock index (heart rate divided by systolic blood pressure), pulse pressure, an oxygen-to-respiratory-rate ratio, and a count of “red flag” vitals (tachypnoea, hypoxia, fever) — on the reasoning that a model should not have to rediscover basic clinical relationships from raw numbers alone.

Three complex model families were built on the engineered features: Random Forest (300 trees, untuned), Random Forest tuned via RandomizedSearchCV (8 candidate hyperparameter combinations × 3-fold cross-validation, 24 forests fitted in total), Gradient Boosting (HistGradientBoostingClassifier, depth 6, 300 boosting rounds), and a small MLP neural network (two hidden layers of 64 and 32 units). Each was benchmarked against both Week 6 baselines — rebuilt fresh on the identical split for a fair comparison — on six quantitative axes (accuracy, precision, recall, F1, training time, inference time) plus a seventh, qualitative one: whether a single prediction can be explained to Dr Reyes in under a minute.

Benchmark Table

Model	Acc.	Macro F1	Recall ESI1	Train time	Infer. (ms/pt)	Interpretability
Week 6 — Logistic Regression	0.667	0.492	0.25	1.8 s	0.007	High — signed coefficients
Week 6 — Decision Tree	0.556	0.216	0.00	0.2 s	0.000	High — single traceable path
Random Forest	0.641	0.390	0.00	38.2 s	0.132	Medium — importances only
Random Forest (tuned)	0.608	0.475	0.31	1,099 s	0.050	Medium
Gradient Boosting	0.550	0.416	0.31	7.7 s	0.007	Medium-Low
Small MLP	0.638	0.498	0.31	103.1 s	0.003	Low — needs SHAP/LIME

The number this memo turns on: three of the four Week 7 models converge on the same ESI-1 recall (0.31) — but the cost of reaching it ranges from 7.7 seconds to over 18 minutes.

Compute Cost, in Plain Terms

For Martina Griffith specifically: 1,099 seconds does not sound like much on its own, but it is the cost of one training run on a single CPU core with no other load. In practice that is roughly 18 minutes of dedicated compute every time the model needs retraining — after a data refresh, a quarterly review, or a change to the feature set — multiplied by however many times the team iterates before a version is judged safe to trial. Gradient Boosting's 7.7 seconds is, for all practical purposes, free: it can be retrained on demand, inside a normal

development loop, without booking time on shared infrastructure or waiting on a background job. That difference compounds every time this project touches the model again, not just once at launch.

Three Arguments For Gradient Boosting

- It matches the best available clinical benefit. Recall on ESI 1 (0.31) is tied with the tuned Random Forest and the MLP, more than double the Week 6 logistic-regression baseline (0.25) — roughly five critical patients caught in the test set instead of four.
- It is dramatically cheaper to produce. Gradient Boosting trains in 7.7 seconds; the tuned Random Forest needed 1,099 seconds (~18 minutes) to reach the identical clinical outcome — for Martina Griffith, that is the difference between a model retrainable weekly on a laptop and one needing dedicated compute time budgeted every refresh, for zero extra benefit.
- Inference is effectively free, and it stays explainable enough to defend. At 0.007 ms per patient it matches logistic regression at deployment time, and feature importances give a workable, if second-tier, explanation — a materially better story for Dr Reyes than the MLP's opaque weights.

Three Arguments Against

- Absolute performance is still nowhere near deployable. A recall of 0.31 for ESI 1 means roughly two in three critical patients are still missed — a difference in degree from the Week 6 baseline, not a solved problem.
- It is the weakest model overall on everyday triage. Gradient Boosting's accuracy (0.550) and macro F1 (0.416) are the lowest of all six models benchmarked. It appears to trade away performance on the common ESI 2–4 cases to gain a little on the rare ESI 1 class — a real trade-off, not a free win.
- Interpretability is a step down from the current baseline. Feature importances explain which inputs mattered on average, not why any individual patient received their prediction — weaker than logistic regression's signed, per-patient coefficients for the one-minute clinician explainer.

Risks & Unknowns

- We do not yet know how far off the misses land. This week's benchmark reports aggregate recall only. The Week 6 report's most concerning failure mode was patients under-triaged by two levels, not one — that breakdown has not been re-run for any Week 7 model.
- No subgroup fairness audit has been done. Random Forest performance shifts when demographic features are added, but no Week 7 model has been checked for uneven performance across demographic groups — a deployment blocker if left unresolved, independent of which model is chosen.

Recommendation

Do not deploy any model built this week. Carry Gradient Boosting forward as the Phase 3 development track over the tuned Random Forest (identical clinical benefit, ~140× the compute cost) and over the MLP (identical clinical benefit, weaker interpretability, ~13× slower to train). Do not pursue further Random Forest hyperparameter search — this week's search spent 18 minutes of compute to land exactly where Gradient Boosting starts for free.

Phase 3 effort should target ESI-1 recall specifically — for example, cost-sensitive class weighting or resampling techniques that penalise missed critical cases directly — rather than further model complexity, since complexity alone has now visibly plateaued at 0.31 recall across three different model families. Before any deployment conversation, the two items in Risks & Unknowns above (near-miss severity, subgroup fairness) should be closed out; a model cannot be called defensible to Dr Reyes or the Board while either remains open.

Reproducibility

All code is committed to GitHub: Week7_Tutorial2_Optimization_Techniques.ipynb in /notebooks (rebuilds both Week 6 baselines and all four Week 7 models on the identical split). Benchmark data in

/docs/week7_benchmark_table.csv. random_state = 42 is fixed throughout, so every figure in this memo is exactly reproducible from the committed notebook.

References

- [1] Agency for Healthcare Research and Quality, *Emergency Severity Index (ESI): A Triage Tool for Emergency Department Care, Version 4 — Implementation Handbook*, 2020 ed. Rockville, MD: AHRQ.
- [2] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [3] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.